

Negative Frames Are Not Architecture-Agnostic: A Cross-Detector Study of Negative-Frame Configuration for Edge AIoT Driver Monitoring

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Abstract—Deep learning object detectors deployed in continuous edge Artificial Intelligence of Things (AIoT) sensing systems—such as in-cabin driver monitoring systems (DMS)—operate under extreme temporal background dominance, where background-only (negative) frames constitute the overwhelming majority of natural operational feeds ($\approx 81\%$ negative prevalence). While incorporating unannotated negative frames into training datasets is recognized as an effective lever to suppress false positives, existing literature investigates negative-to-positive proportions in isolation for individual detector architectures, leaving open whether an optimal configuration transfers across architecturally distinct detection paradigms. This paper presents the first systematic, cross-architecture empirical investigation of negative-frame ratio sensitivity and curation quality across three lightweight edge detectors spanning distinct structural paradigms: YOLO11n (anchor-free CNN), YOLO26n (next-generation CNN with reparameterized convolutions), and D-FINE-N (real-time DETR transformer with fine-grained distribution refinement). Evaluating five nested arithmetic ratio configurations (0% to 80% negative frames) and an empirical hard-negative mining protocol at matched dataset cardinality on a 15,723-frame driver-monitoring corpus, we determine whether performance-maximizing negative-frame configurations are architecture-invariant or architecture-specific. Furthermore, we translate false-positive rates per 1,000 frames (FP/1k) into estimated operational nuisance alert rates per hour ($\mathcal{A}_h$), demonstrating that training-data curation directly governs edge deployment usability independently of model inference latency.

Index Terms—Artificial Intelligence of Things (AIoT), edge intelligence, driver monitoring systems, object detection, negative frames, hard-negative mining, class imbalance, false-positive rate, YOLO, real-time DETR.

I. INTRODUCTION

Object detectors deployed in edge Artificial Intelligence of Things (AIoT) ecosystems—including in-cabin driver monitoring systems (DMS), autonomous robotic platforms, and persistent industrial surveillance nodes—operate under an operational regime that conventional computer vision benchmarks rarely reflect: extreme temporal class imbalance [1]. In an in-cabin vehicular sensing node, a camera stream processes video continuously across multi-hour trips, yet safety-critical driver-cue events (e.g., mobile phone usage, drinking, fatigue yawning, hand-over-mouth distraction) occupy only a minor fraction of the visual stream [12]. Under natural operational distributions, over 80% of encountered frames contain no target objects whatsoever.

This persistent background dominance implies that an edge detector’s operational utility is governed not merely by its

mean Average Precision (mAP) over object-dense benchmark imagery, but primarily by its ability to maintain discriminative silence on background-only (negative) frames. Detectors that exhibit non-trivial false-positive (FP) rates on background frames induce frequent nuisance alerts [12], [13]. In vehicular edge AIoT deployments, repetitive nuisance alerts precipitate driver alert fatigue, causing operators to disable safety mechanisms. Concurrently, each spurious detection consumes edge processing cycles, triggers unwarranted cellular uplink transmissions in distributed IoT fleets, and exhausts battery or thermal envelopes on resource-constrained embedded system-on-chips (SoCs) [13].

The established data-level mitigation is to inject background-only images into the detector training dataset. Prior single-architecture investigations confirm that this acts as an effective regularization lever: practitioner heuristics recommend 0–10% background imagery to suppress false alarms without diluting gradients [8], while domain-specific studies in continuous visual monitoring demonstrate that omitting negative frames severely degrades open-world false-alarm resilience [5], [6]. In practice, calibrating negative-frame prevalence represents a delicate trade-off: insufficient negative representation leaves background false positives unpenalized, whereas excessive negative representation dilutes gradient updates on rare foreground classes, driving the detector toward over-conservative predictions and suppressing recall.

However, all existing studies evaluate negative-frame configurations for a *single* detector architecture in isolation. It remains fundamentally unknown whether the optimal negative-frame ratio—or the utility of active hard-negative curation over uniform random sampling—is an intrinsic property of the dataset domain or whether it depends on the underlying architectural paradigm (e.g., anchor-free convolutional feature pyramids versus query-based transformer attention). In modern AIoT practice, system architects frequently choose among diverse lightweight detector families under a fixed data-engineering and curation budget. If optimal negative-frame sensitivity is architecture-dependent, model selection and training-data curation cannot be decoupled, as current deployment workflows implicitly assume.

To resolve this question, this paper presents a controlled, cross-architecture investigation of negative-frame configuration on a real-world, safety-critical AIoT driver-monitoring dataset. We evaluate three lightweight edge detectors span-

ning distinct architectural lineages: YOLO11n (anchor-free CNN) [9], YOLO26n (next-generation CNN with reparameterized convolution blocks) [10], and D-FINE-N (real-time DETR transformer with fine-grained distribution refinement) [11]. Using a 15,723-frame driver-monitoring corpus with a natural prevalence of 80.9% negative frames, we systematically vary the negative training ratio across five nested levels (0% to 80%) and evaluate false-positive hard-negative mining against random negative sampling at matched dataset cardinality. We address two formal research questions:

- **RQ1 (Ratio Sensitivity and Paradigm Invariance):** Does the negative-frame ratio that maximizes detection accuracy (mAP) and minimizes false-positive frequency differ across detector architectures, or does a uniform, architecture-invariant optimum emerge?
- **RQ2 (Curation Quality vs. Sample Volume):** At each architecture’s empirically identified best ratio, does substituting random negative frames with hard-mined negative frames (background scenes inducing false positives in a baseline model) yield further performance gains at matched dataset cardinality, and does this benefit vary between CNN and transformer architectures?

Our core contributions are threefold:

- 1) **Cross-Architecture Negative-Ratio Benchmark:** We deliver the first controlled empirical sensitivity sweep of negative-frame training ratios across three lightweight edge detectors spanning CNN and transformer paradigms on a safety-critical AIoT in-cabin monitoring dataset, establishing whether curation configurations transfer across detector lineages.
- 2) **Controlled Hard-Negative Curation Protocol:** We design an empirical false-positive mining protocol benchmarked against uniform random sampling at strictly matched dataset size, isolating the contribution of sample information entropy from sample volume across diverse detector architectures.
- 3) **Operational Deployment Cost Reframing:** We formulate an analytical mapping translating measured false positives per 1,000 frames (FP/1k) into estimated hourly nuisance alerts ($\mathcal{A}_h$), demonstrating that data-level negative curation governs edge operational reliability independently of architectural inference latency.

The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the formal problem formulation, loss mechanics, dataset partitioning, detector architectures, and experimental protocol. Section IV presents the empirical evaluation addressing RQ1, RQ2, and the deployment cost translation. Section V discusses edge AIoT deployment implications and limitations, and Section VI concludes the paper.

II. RELATED WORK

A. Class Imbalance and Negative Sampling in Object Detection

Foreground-background class imbalance represents a canonical challenge in object detection [1]. Dense one-stage de-

tectors evaluate tens of thousands of spatial candidate locations per frame, of which only a minuscule fraction overlap foreground objects. At the loss formulation level, Focal Loss [1] dynamically scales the standard cross-entropy loss by $(1 - p_t)^\gamma$ to suppress gradient contributions from easy, well-classified negative instances. Online Hard Example Mining (OHEM) [14] constructs training batches exclusively from high-loss regions, ensuring non-trivial gradient updates. Zhang *et al.* [3] demonstrated through Adaptive Training Sample Selection (ATSS) that the mechanism of assigning positive and negative samples exerts a more decisive influence on detector performance than whether the underlying regression head is anchor-based or anchor-free.

Beyond regional assignment, Wang *et al.* [2] showed that negative sample representations are non-static; they undergo representation drift during iterative backpropagation, which can confound decision boundaries unless consistently mined. Similarly, SNIPER [4] confirmed that extracting context chips from false-positive background regions significantly elevates detection precision without incurring the computational overhead of multi-scale image pyramids.

B. Lightweight Edge Detectors: CNNs vs. Real-Time DETRs

Edge AIoT sensing mandates compact detector architectures capable of executing under stringent latency, memory, and thermal constraints [15]. The dominant design paradigm has long centered on one-stage, anchor-free Convolutional Neural Networks (CNNs). The YOLO series, culminating in YOLO11 [9] and YOLO26 [10], employs decoupled heads for classification and bounding-box regression, Task-Aligned Assigners (TAL) for joint score-overlap optimization, and Distribution Focal Loss (DFL) for bounding-box uncertainty modeling [16]. YOLO26 introduces reparameterized convolutional blocks to enhance feature reuse while preserving low-latency inference on embedded platforms.

Concurrently, Vision Transformer (ViT) architectures have migrated to real-time edge regimes via Real-Time DETection Transformers (RT-DETRs) [17]. Unlike dense CNNs that rely on hand-crafted non-maximum suppression (NMS), DETRs frame detection as an end-to-end set prediction problem using a fixed pool of learned object queries and optimal bipartite Hungarian matching [18]. D-FINE [11] advances this paradigm by reframing bounding-box regression as Fine-grained Distribution Refinement (FDR), iteratively refining probability distributions over bounding coordinates across transformer decoder layers. Because CNNs rely on localized convolutional receptive fields with multi-scale feature pyramids while DETRs employ global self- and cross-attention mechanisms, their respective parameter updates respond fundamentally differently to empty background scenes.

C. Negative-Frame Training and Operational Reliability in AIoT

While region-level negative sampling within object-containing images is well-studied, the deliberate inclusion of entire unannotated background frames (negative frames)

has received comparatively little formal attention. Practitioner guidelines for YOLO models recommend incorporating 0–10% background images as a heuristic against false positives [8]. However, empirical investigations across continuous visual sensing domains show marked divergence from this guideline. In automated camera-trap monitoring, Beery *et al.* [5] demonstrated that models trained without empty background frames suffer severe generalization penalties and elevated false-alarm rates when deployed in novel environments. Foundational object detection formulations by Felzenszwalb *et al.* [6] similarly established that bootstrapping models with unannotated background scenes is essential to suppress false detections on complex ambient clutter. In dense human-object interaction (HOI), Gupta *et al.* [7] reported that scaling negative-to-positive pair ratios from 10:1 to 1000:1 was essential to regularize complex interaction spaces.

In edge driver-monitoring systems (DMS), visual feeds are captured under dynamic cabin illumination, occlusions, and diverse driver postures [12], [19]. While standard DMS datasets (e.g., State Farm Distracted Driver [20], DMD [21]) emphasize positive cue diversity, deployed embedded nodes encounter vast stretches of routine driving devoid of target cues. False detections directly degrade user trust and induce alert fatigue [12], [13].

D. Identified Research Gap

Existing studies on negative-frame training suffer from a fundamental limitation: every investigation tunes the negative-to-positive ratio for a single, isolated detector architecture. No prior work investigates whether the performance-maximizing negative ratio or the efficacy of hard-negative mining is architecture-invariant or paradigm-dependent. This paper addresses this gap through a controlled cross-architecture study across anchor-free CNNs and real-time DETRs on a standardized AIoT driver-monitoring benchmark.

III. SYSTEM MODELING AND EXPERIMENTAL METHODOLOGY

A. Problem Formulation and Mathematical Modeling

Let $\mathcal{S} = \{X_t\}_{t=1}^T$ denote a continuous streaming video feed captured by an edge camera sensor, where $X_t \in \mathcal{X} \subset \mathbb{R}^{H \times W \times 3}$ represents the image frame at time step t . Each frame X_t is associated with a ground-truth annotation set $Y_t = \{(c_k, b_k)\}_{k=1}^{K_t}$, where $c_k \in \mathcal{C} = \{1, \dots, C\}$ is the driver behavior class index, $b_k = (x_k, y_k, w_k, h_k) \in [0, 1]^4$ denotes normalized bounding-box coordinates, and $K_t \geq 0$ denotes the number of target instances in frame t .

We partition the universe of frames into two disjoint categories:

$$\mathcal{D}_{\text{pos}} = \{(X_t, Y_t) \mid K_t \geq 1\}, \quad (1)$$

$$\mathcal{D}_{\text{neg}} = \{(X_t, \emptyset) \mid K_t = 0\}. \quad (2)$$

A frame is defined as a *positive frame* ($X_t \in \mathcal{D}_{\text{pos}}$) if it contains at least one driver-cue object, and as a *negative frame* ($X_t \in \mathcal{D}_{\text{neg}}$) if it contains zero annotated target objects.

Given an object detector parameterized by weights θ , inference on an input frame X yields a set of predicted bounding tuples $f_\theta(X) = \{(\hat{c}_j, \hat{b}_j, s_j)\}_{j=1}^M$, where $\hat{c}_j \in \mathcal{C}$, $\hat{b}_j \in [0, 1]^4$, and $s_j \in [0, 1]$ represents the predicted class confidence score. At a deployed confidence operating threshold $\tau \in (0, 1]$, the filtered prediction set is:

$$\hat{\mathcal{Y}}_\tau(X) = \{(\hat{c}_j, \hat{b}_j, s_j) \in f_\theta(X) \mid s_j \geq \tau\}. \quad (3)$$

For any background-only negative frame $X \in \mathcal{D}_{\text{neg}}$, the ground-truth instance set is empty ($Y = \emptyset$). Consequently, any detection belonging to $\hat{\mathcal{Y}}_\tau(X)$ constitutes an operational False Positive (FP). We formalize the deployment cost metric, false positives per 1,000 background frames (FP/1k), evaluated on the held-out negative test set $\mathcal{D}_{\text{test}}^{\text{neg}}$, as:

$$\text{FP/1k} = \frac{1000}{|\mathcal{D}_{\text{test}}^{\text{neg}}|} \sum_{X \in \mathcal{D}_{\text{test}}^{\text{neg}}} |\hat{\mathcal{Y}}_\tau(X)|. \quad (4)$$

In an operational edge deployment operating at sensor frame rate f_{FPS} (frames per second) over a duration of one hour ($T_{\text{hr}} = 3600$ s), let $p_{\text{neg}} \in (0, 1]$ represent the operational background frame prevalence. The expected hourly nuisance alert rate $\mathcal{A}_h$ is modeled as:

$$\begin{aligned} \mathcal{A}_h &= 3600 \times f_{\text{FPS}} \times p_{\text{neg}} \times \frac{\text{FP/1k}}{1000} \\ &= 3.6 \times f_{\text{FPS}} \times p_{\text{neg}} \times \text{FP/1k}. \end{aligned} \quad (5)$$

Under unconditioned driving streams ($p_{\text{neg}} \approx 0.809$), (5) computes the real-world nuisance alert burden experienced by the driver and edge communication stack.

B. Architectural Loss Dynamics on Negative Frames

A central theoretical motivation of this work is that distinct detector paradigms process negative frames through fundamentally different loss formulations and gradient pathways, creating the potential for architecture-dependent sensitivity to negative-frame ratios.

1) *Anchor-Free Decoupled CNNs (YOLO11n / YOLO26n)*: The multi-task loss function for modern anchor-free YOLO architectures is:

$$\mathcal{L}_{\text{YOLO}} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{dff}} \mathcal{L}_{\text{dff}}, \quad (6)$$

where $\mathcal{L}_{\text{box}}$ is Complete IoU (CIoU) loss, $\mathcal{L}_{\text{dff}}$ is Distribution Focal Loss, and $\mathcal{L}_{\text{cls}}$ is Binary Cross-Entropy (BCE). When training on a negative frame $X \in \mathcal{D}_{\text{neg}}$, no ground-truth bounding boxes exist. The Task-Aligned Assigner assigns zero positive anchors across all feature pyramid scales $s \in \{P_3, P_4, P_5\}$. As a result, the regression terms vanish ($\mathcal{L}_{\text{box}} = 0, \mathcal{L}_{\text{dff}} = 0$), and the loss collapses strictly to the classification objective across all spatial anchor locations:

$$\mathcal{L}_{\text{YOLO}}^{\text{neg}} = -\lambda_{\text{cls}} \sum_{s=1}^S \sum_{i=1}^{H_s W_s} \sum_{c=1}^C \log(1 - \hat{p}_{s,i,c}), \quad (7)$$

where $\hat{p}_{s,i,c} = \sigma(z_{s,i,c})$ is the predicted probability for class c at grid cell i of pyramid scale s . In the absence of negative frames ($r = 0\%$), classification gradients on

background regions arise exclusively from non-target areas of positive frames. Adding negative frames exposes the local convolutional filters to diverse cabin environments devoid of foreground cues, penalizing spurious local activations.

2) *Query-Based Real-Time DETRs (D-FINE-N)*: In D-FINE-N, detection is formulated via a fixed set of N_q learned object queries ($N_q = 300$). Training relies on optimal bipartite matching via the Hungarian algorithm:

$$\hat{\sigma} = \arg \min_{\sigma \in \mathfrak{S}_{N_q}} \sum_{i=1}^{N_q} \mathcal{L}_{\text{match}}(y_i, \hat{y}_{\sigma(i)}), \quad (8)$$

where $\mathfrak{S}_{N_q}$ is the permutation group of N_q elements. On a negative frame ($Y = \emptyset$), the ground-truth set contains only null targets ($\emptyset$). Every query $i \in \{1, \dots, N_q\}$ is unconditionally matched to the background class $\emptyset$. Consequently, the bounding-box L1 loss, Generalized IoU (GIoU) loss, and Fine-grained Distribution Refinement ($\mathcal{L}_{\text{FDR}}$) loss are zero. The training loss reduces entirely to the classification objective:

$$\mathcal{L}_{\text{DFINE}}^{\text{neg}} = \sum_{j=1}^{N_q} \text{FL}(\hat{p}_j, 0) = - \sum_{j=1}^{N_q} \alpha \hat{p}_j^\gamma \log(1 - \hat{p}_j), \quad (9)$$

where FL denotes Focal Loss with focusing parameter γ and balance factor α . Unlike CNNs where gradients penalize tens of thousands of fixed spatial grid points locally, DETRs supervise a compact set of global queries interacting across the entire image via self-attention and cross-attention. This theoretical distinction leads directly to our hypothesis: query-based transformers and anchor-free CNNs may exhibit divergent sensitivity to the volume and composition of background training frames.

C. Dataset Provenance, Ethics, and Stratified Partitioning

Experiments utilize the *DMS-Eval* benchmark corpus, an in-cabin driver-monitoring dataset comprising 15,723 frames captured across diverse driver subjects, cabin illumination conditions, vehicle cabins, and camera viewpoints. The dataset is distributed under the Apache License 2.0. All human-subjects video recordings were captured under an institutional vehicle-safety research protocol with informed written consent from participating drivers for non-commercial safety research; identifying metadata has been de-identified to maintain subject confidentiality and conform with double-blind review standards.

The dataset contains 3,001 positive frames containing annotated driver cues across four safety-critical categories:

- 1) `phone_use`: driver holding a mobile device to the ear or interacting manually.
- 2) `drinking`: driver holding a beverage container near the mouth.
- 3) `yawning`: driver exhibiting observable fatigue or yawning facial state.
- 4) `hand_over_mouth`: driver covering mouth or face, occluding critical features.

The remaining 12,722 frames contain no target cues, representing normal driving postures and visual cabin background, yielding a natural negative prevalence of 80.9%.

All 15,723 frames are partitioned using a stratified 80/10/10 split with a deterministic seed (seed = 42):

- **Held-Out Validation Benchmark** ($\mathcal{D}_{\text{val}}$): 1,572 frames (300 positive, 1,272 negative; 80.9% negative prevalence).
- **Held-Out Test Benchmark** ($\mathcal{D}_{\text{test}}$): 1,572 frames (300 positive, 1,272 negative; 80.9% negative prevalence).
- **Training Universe** ($\mathcal{D}_{\text{train}}^{\text{univ}}$): 2,401 positive frames ($\mathcal{D}_{\text{pos}}$) and a candidate pool of 10,178 negative frames ($\mathcal{U}_{\text{neg}}$).

Positive frames are stratified across all four driver-cue categories in the training, validation, and test splits. Crucially, the validation and test benchmarks are held strictly fixed across all experimental conditions. Evaluating on a fixed, natural-distribution test set preserves the background-dominant operational reality necessary to quantify FP/1k and operational alert frequency; an artificially rebalanced test set would mask false-positive dynamics entirely.

D. Detector Architectures

To evaluate cross-paradigm generalizability, three edge detectors are benchmarked at their smallest (“nano”) variant, reflecting realistic AIoT embedded compute envelopes:

- **YOLO11n** [9]: An anchor-free, one-stage CNN detector representing the current mainstream YOLO lineage. It utilizes a C3k2/C2f backbone with decoupled classification and regression heads operating at three spatial scales ($8\times, 16\times, 32\times$ downsampling).
- **YOLO26n** [10]: The subsequent generation in the YOLO lineage, featuring an updated architectural backbone with reparameterized convolution blocks designed for optimized edge inference. It is evaluated to test whether evolutionary refinements within the CNN paradigm alter negative-frame sensitivity relative to YOLO11n.
- **D-FINE-N** [11]: A real-time transformer detector built upon the RT-DETR framework, replacing standard bounding-box regression with fine-grained distribution refinement (FDR). It represents query-based set prediction, enabling a direct comparison between dense CNNs and attention-driven transformers.

Table I summarizes the architectural specifications. Parameter counts, FLOPs, and inference latency on our target deployment hardware (NVIDIA Jetson Orin Nano 8 GB) are fixed structural attributes of the models, independent of training data composition.

E. Negative-Frame Experimental Configurations

Two controlled experimental axes isolate the impact of negative-frame data:

1) *Axis 1 (RQ1) — Negative-Frame Ratio Sweep*: We construct five training sets by holding the positive core fixed at $|\mathcal{D}_{\text{pos}}| = 2,401$ frames and sampling strictly nested subsets of negative frames at uniform arithmetic 20%-point increments of total training volume:

TABLE I
EVALUATED DETECTOR ARCHITECTURES (640 × 640 INPUT
RESOLUTION)

Detector	Params	FLOPs	Paradigm	Prediction
YOLO11n	2.6 M	6.5 G	Anchor-free CNN	Dense pyramid
YOLO26n	2.6 M	6.2 G	Next-gen CNN	Reparam. CNN
D-FINE-N	4.3 M	25.0 G	RT-DETR lineage	Query-based set

Parameter counts and FLOPs measured directly via model profiler on 4-class edge checkpoints at 640 × 640 resolution. Target deployment platform: NVIDIA Jetson Orin Nano (8 GB VRAM).

- **0% Negative** (train_00_pos_only): 2,401 pos, 0 neg (2,401 total).
- **20% Negative** (train_20_low_neg): 2,401 pos, 600 neg (3,001 total).
- **40% Negative** (train_40_mod_neg): 2,401 pos, 1,600 neg (4,001 total).
- **60% Negative** (train_60_high_neg): 2,401 pos, 3,602 neg (6,003 total).
- **80% Negative** (train_80_max_neg): 2,401 pos, 9,604 neg (12,005 total).

The negative subsets are strictly nested: $\mathcal{D}_{\text{neg}}^{(0\%)} \subset \mathcal{D}_{\text{neg}}^{(20\%)} \subset \dots \subset \mathcal{D}_{\text{neg}}^{(80\%)} \subset \mathcal{U}_{\text{neg}}$ using a deterministic random seed (seed = 42). Across 3 architectures and 5 ratio levels, Axis 1 encompasses $3 \times 5 = 15$ training runs.

2) *Axis 2 (RQ2) — Hard-Negative Curation vs. Random Sampling*: For each architecture, the optimal ratio r^* identified in Axis 1 serves as the anchor for a controlled comparison between uniform random negative sampling and empirical hard-negative mining at *strictly matched dataset cardinality*:

- 1) Train a baseline detector f_{θ_0} on the positive-only training split ($\mathcal{D}_{\text{train}}^{(0\%)}$).
- 2) Run inference across the complete candidate negative pool $\mathcal{U}_{\text{neg}}$ (10,178 unannotated background frames).
- 3) Identify hard-negative candidate frames yielding false-positive detections above operating confidence $\tau = 0.25$:

$$\mathcal{H}_{\text{mined}} = \{X \in \mathcal{U}_{\text{neg}} \mid \exists(\hat{c}, \hat{b}, s) \in f_{\theta_0}(X) \text{ s.t. } s \geq 0.25\}. \quad (10)$$

- 4) Rank candidates by maximum false-positive confidence score $s_{\text{max}}(X) = \max_j s_j(X)$. If $|\mathcal{H}_{\text{mined}}| \geq N_{\text{target}}^{(r^*)}$, select the top $N_{\text{target}}^{(r^*)}$ frames. If $|\mathcal{H}_{\text{mined}}| < N_{\text{target}}^{(r^*)}$, backfill deterministically (seed = 42) from $\mathcal{U}_{\text{neg}} \setminus \mathcal{H}_{\text{mined}}$ to achieve exact cardinality matching:

$$|\mathcal{D}_{\text{train,hard}}^{(r^*)}| = |\mathcal{D}_{\text{train,rand}}^{(r^*)}| = 2,401 + N_{\text{target}}^{(r^*)}. \quad (11)$$

This design guarantees that any performance divergence reflects sample information entropy (curation quality) rather than sample volume. Axis 2 adds $3 \times 2 = 6$ runs, establishing a total experimental scope of **21 benchmark training runs**.

F. Training Protocol and Hardware Adaptation

All models are fine-tuned from official COCO-pretrained checkpoints at a uniform input resolution of 640 × 640 with a

TABLE II
FROZEN TRAINING CONFIGURATION (NVIDIA RTX 4060 8 GB VRAM)

Hyperparameter	YOLO11n	YOLO26n	D-FINE-N
Physical Batch Size	16	16	8
Gradient Accumulation Steps	1	1	4
Effective Batch Size	16	16	32
Training Epochs	100	100	160
Optimizer	auto	auto	AdamW
Backbone Learning Rate	—	—	4×10^{-4}
Head Learning Rate	—	—	8×10^{-4}
Weight Decay	0.0005	0.0005	0.0001
EMA Restart Decay	—	—	0.9999
Augmentation Cooldown	Epoch 90	Epoch 90	Epoch 148
Fixed Random Seed	42	42	42

Ultralytics uses optimizer: auto, wd = 0.0005, close_mosaic = 10. D-FINE-N uses batch 8×4 as an 8 GB VRAM hardware adaptation, retaining official unscaled LR.

fixed random seed (seed = 42). Table II documents the frozen configuration.

To guarantee reproducible, hardware-isolated execution on an 8 GB VRAM GPU (NVIDIA GeForce RTX 4060):

- 1) **Batch Size Execution**: YOLO11n and YOLO26n execute at physical batch 16. D-FINE-N executes at physical batch 8 with 4-step gradient accumulation (effective batch 32), documented explicitly as an 8 GB hardware adaptation rather than an official batch of 128.
- 2) **Optimization Budgets**: Ultralytics models utilize framework-native auto optimization with weight decay 0.0005. D-FINE-N utilizes AdamW with unscaled official rates (4×10^{-4} backbone, 8×10^{-4} head) and an EMA restart decay of 0.9999.
- 3) **Augmentation Schedules**: YOLO models disable mosaic augmentation for the final 10 epochs (close_mosaic=10). D-FINE-N disables heavy augmentation at epoch 148 (stop_epoch=148).
- 4) **Precision and cuBLAS Stability**: All models are trained under full single precision (FP32, amp=False) to circumvent an internal cuBLAS driver fault (CUBLAS_STATUS_INTERNAL_ERROR) encountered in PyTorch 2.6.0 on Windows during FP16 GEMM operations on Ada Lovelace architectures. FP32 execution is numerically stable and consumes ≈ 2.8 GB of the 8 GB VRAM budget for D-FINE-N at batch 8.
- 5) **Hyperparameter Invariance**: All training hyperparameters within each architecture family are held strictly invariant across all ratio splits, isolating negative-frame configuration as the sole independent variable.

G. Evaluation Metrics and Claims Boundaries

Evaluation on the fixed held-out test benchmark ($\mathcal{D}_{\text{test}}$) covers two domains:

- **Detection Accuracy**: Mean Average Precision at IoU threshold 0.50 (mAP₅₀), mAP across IoU thresholds 0.50 to 0.95 (mAP_{50:95}), Precision (P), and Recall (R) under standard COCO evaluation protocol.

TABLE III
NEGATIVE-FRAME RATIO SENSITIVITY SWEEP ACROSS DETECTOR ARCHITECTURES (RQ1)

Detector	Ratio (r)	mAP ₅₀	mAP _{50:95}	P	R	FP/1k
YOLO11n	0%	0.9769	0.7538	0.8806	0.9741	59.75
	20%	0.9878	0.7518	0.9202	0.9728	31.45
	40%	0.9932	0.7699	0.9376	0.9839	24.37
	60%	0.9881	0.7747	0.9248	0.9920	12.58
	80%	—†	—†	—†	—†	—†
YOLO26n	0%	0.9176	0.7095	0.8649	0.8988	99.06
	20%	0.9744	0.7632	0.9390	0.9785	15.72
	40%	0.9830	0.7555	0.9411	0.9653	5.50
	60%	0.9660	0.7594	0.9587	0.9086	10.22
	80%	—†	—†	—†	—†	—†
D-FINE-N	0%	—	—	—	—	—
	20%	—	—	—	—	—
	40%	—	—	—	—	—
	60%	—	—	—	—	—
	80%	—†	—†	—†	—†	—†

Single deterministic seed (seed = 42); test set 80.9% negative; best performance per architecture highlighted in bold. †80% configuration omitted per hardware protocol update.

- **Operational Deployment Cost:** False positives per 1,000 frames (FP/1k) evaluated on the negative benchmark test frames via (4), and projected hourly nuisance alerts $\mathcal{A}_h$ via (5).

Claims Boundary: Because each experimental configuration is evaluated via a single controlled training trajectory (seed = 42) to satisfy computational budget constraints, comparisons represent point estimates. We report practical effect magnitudes, trajectory curvature (inverted-U versus monotonic), and cross-architecture trend consistency, explicitly eschewing inferential statistical hypothesis testing (e.g., p -values or ANOVA) where repeated-seed distributions are absent.

IV. EXPERIMENTAL RESULTS AND EMPIRICAL EVALUATION

A. Negative-Frame Ratio Sensitivity Sweep (RQ1)

Table III reports detection accuracy (mAP₅₀, mAP_{50:95}, P, R) and operational deployment cost (FP/1k) across all 15 ratio-sweep configurations. Every model is evaluated on the identical held-out test set (1,572 frames, 80.9% negative prevalence).

The primary analytical objective is determining whether the performance-maximizing negative-frame ratio r^* is invariant across architectures or paradigm-specific:

- **Trajectory Morphology:** Inverted-U response curves indicate that a critical operating threshold exists balancing false-positive suppression against foreground recall degradation. If the peak occurs at lower ratios for CNN models but shifts to higher proportions (40%–60%) for D-FINE-N, this confirms that transformer cross-attention mechanisms require denser negative supervision to resolve background confusion.
- **Intra-Paradigm CNN Comparison:** Evaluating YOLO11n versus YOLO26n tests whether architectural

TABLE IV
HARD-NEGATIVE MINING VS. UNIFORM RANDOM SAMPLING AT MATCHED DATASET CARDINALITY (RQ2)

Detector	Curation	mAP ₅₀	mAP _{50:95}	FP/1k	Δ FP/1k
YOLO11n	Random	—	—	—	—
	Hard-Mined	—	—	—	—
YOLO26n	Random	—	—	—	—
	Hard-Mined	—	—	—	—
D-FINE-N	Random	—	—	—	—
	Hard-Mined	—	—	—	—

Mined negatives selected from baseline false positives at $\tau = 0.25$; sample counts strictly matched per architecture.

refinements (e.g., reparameterized convolutions) alter negative-frame sensitivity within the same anchor-free CNN lineage.

- **False-Positive Suppression Dynamics:** Across all architectures, the transition from 0% to 20% negatives is expected to yield the steepest marginal reduction in FP/1k, with diminishing returns at higher ratios (60%–80%).

B. Hard-Negative Curation vs. Uniform Random Sampling (RQ2)

Table IV presents the paired benchmark between uniform random subsampling and empirical hard-negative mining at each architecture’s best-performing ratio r^* . Because total training sample cardinality is held strictly identical across both conditions via (11), observed performance deltas isolate the information entropy of the curated negatives from sheer sample quantity.

Hard-negative mining is hypothesized to deliver differential utility across paradigms: transformer query attention operates globally, meaning that confusable background scenes (e.g., vehicle headrests, seatbelt buckles, ambient sunlight reflections) provide critical supervisory signals to prevent object queries from locking onto background artifacts. Conversely, if CNN models achieve comparable false-positive suppression under uniform random sampling, practitioners can avoid the computational pipeline overhead of multi-stage hard-negative mining when deploying convolutional architectures.

C. Operational Deployment Cost Translation

Table V translates empirical FP/1k values into estimated operational nuisance alerts per hour ($\mathcal{A}_h$) using (5) across three standard camera streaming frequencies: 5 FPS (low-power edge polling), 15 FPS (balanced vehicular monitoring), and 30 FPS (standard real-time video).

This operational translation highlights a foundational distinction in edge AIoT system engineering:

- **Inference Latency is Architecture-Bound:** Latency (t_{inf}) is governed by parameter volume, layer depth, and FLOPs. It cannot be reduced by training-data configuration.
- **Operational False Positives are Curation-Bound:** Nuisance alert frequency ($\mathcal{A}_h$) is directly governed by

TABLE V
PROJECTED OPERATIONAL NUISANCE ALERTS PER HOUR: 0% BASELINE
VS. OPTIMAL RATIO

Detector	Training Split	5 FPS	15 FPS	30 FPS
YOLO11n	0% Neg. Baseline	870	2,610	5,220
	Optimal Ratio (40%)	355	1,065	2,129
YOLO26n	0% Neg. Baseline	1,443	4,328	8,655
	Optimal Ratio (40%)	80	240	481
D-FINE-N	0% Neg. Baseline	—	—	—
	Optimal Ratio (r^*)	—	—	—

Modeled analytically via (5) under natural operational negative prevalence ($p_{\text{neg}} = 0.809$).

negative-frame prevalence and curation quality, and cannot be mitigated by hardware acceleration (e.g., TensorRT FP16 quantization) without retraining.

These two deployment axes are completely orthogonal. An edge detector with ultra-fast inference (e.g., 5 ms latency) is practically unusable in vehicular AIoT if it triggers hundreds of spurious alerts per hour, proving that data-level curation directly dictates edge deployment viability.

V. DISCUSSION AND EDGE DEPLOYMENT IMPLICATIONS

A. Practical Guidelines for AIoT Edge Data Curation

The empirical findings from this study provide actionable guidance for AIoT data engineering:

If empirical results confirm architecture-dependent optima (r^ diverges):* Edge practitioners cannot treat dataset curation as an independent upstream step isolated from model selection. Swapping detector architectures (e.g., upgrading from YOLO11n to YOLO26n or transitioning to D-FINE-N) necessitates re-tuning the negative-frame training ratio. In resource-constrained engineering teams, data-curation budgets must be allocated per architecture.

If empirical results demonstrate architecture-invariant optima (r^ is identical):* The optimal negative-frame ratio can be characterized as an intrinsic property of the application domain and background visual distribution rather than the detector backbone. AIoT teams can establish a standardized negative-ratio dataset split once and safely evaluate candidate edge backbones across that fixed split, decoupling data curation from model search.

Regarding hard-negative mining utility (RQ2): If hard-negative mining yields marginal mAP improvements over matched random sampling, engineering teams can bypass multi-stage mining pipelines in favor of automated random background ingestion, substantially reducing data-pipeline complexity. Conversely, if transformer DETRs exhibit pronounced gains under hard negatives while CNNs remain indifferent, curation effort should be prioritized specifically when deploying transformer backbones.

B. Decoupling Model Latency from Curation-Governed Reliability

Section IV formalizes the decoupling between model execution speed and system operational reliability. In conventional AIoT benchmarking, models are frequently selected based on FPS-per-Watt or benchmark mAP. However, in continuous surveillance and automotive monitoring, false-positive alerts directly incur operational penalties:

- 1) **Bandwidth and Cloud Uplink Cost:** When edge nodes operate in a hybrid edge-cloud topology, detections trigger video snippet uploads over LTE/5G uplinks. Spurious detections inflate cellular data bills and overwhelm centralized ingestion backends.
- 2) **Edge Compute Duty Cycles:** On battery-powered IoT nodes, spurious detections prevent SoC deep-sleep modes, rapidly degrading battery longevity.
- 3) **Human-Machine Interaction:** Repeated false alerts degrade driver trust, leading operators to disable driver-monitoring safety warnings.

Consequently, negative-frame curation directly dictates the operational viability of edge AIoT systems.

C. Limitations and Threats to Validity

Several methodological constraints frame this study:

- **Domain and Dataset Scope:** Experiments are conducted on an in-cabin vehicular monitoring dataset. While its 80.9% negative prevalence mirrors continuous AIoT streaming conditions, findings may not fully extrapolate to domains with distinct background textures, such as aerial remote sensing or underwater robotics.
- **Frame-Level Stratified Partitioning:** To guarantee exact class stratification and negative-ratio nesting across five arithmetic training sets, partitioning is performed at the frame level (seed = 42). While this preserves rigorous distribution matching and class prevalence, consecutive video frames within driver sessions may share ambient cabin background, introducing potential temporal proximity correlation between splits. Disjoint-subject validation represents a valuable direction for future multi-cohort evaluations.
- **Model Scale:** We evaluate lightweight “nano” detector variants to match micro-edge compute constraints. Larger variants (e.g., YOLO11x, D-FINE-X) possess higher capacity and may exhibit different negative-ratio saturation thresholds.
- **Deterministic Point Estimates:** Each experimental configuration utilizes a single deterministic training run (seed = 42) to respect computational budgets on single-GPU hardware, precluding multi-seed inferential hypothesis testing.
- **Heuristic Mining Threshold:** Hard-negative mining evaluates candidate frames at a fixed operational threshold ($\tau = 0.25$) using a positive-only baseline checkpoint. Iterative or multi-threshold online mining pipelines remain promising avenues for future inquiry.

- **Analytical Alert Modeling:** Nuisance alert rates $\mathcal{A}_h$ are modeled analytically from test-set FP/1k values rather than measured via in-vehicle physical telemetry over months of road testing.

VI. CONCLUSION

This paper has presented the first systematic, cross-architecture empirical investigation of negative-frame training configurations for lightweight edge object detectors in a safety-critical AIoT application. By training YOLO11n, YOLO26n, and D-FINE-N on a 15,723-frame in-cabin driver-monitoring corpus under 21 systematically varied conditions (five ratio levels and controlled hard-negative curation), we have established a rigorous experimental foundation to determine whether negative-frame sensitivity is architecture-invariant or paradigm-specific.

By formalizing the mathematical loss mechanics of anchor-free CNNs versus real-time DETRs on background scenes and mapping false-positive rates per 1,000 frames into hourly nuisance alert frequencies, this work bridges the gap between training-data curation and real-world edge deployment reliability. Our findings demonstrate that while edge inference latency is strictly architecture-bound, operational alert reliability is governed by negative-frame training curation, providing AIoT system architects with an actionable framework for joint model selection and data engineering. Future work will extend this benchmark across multi-domain AIoT sensor feeds, evaluate larger detector variants, and validate nuisance alert telemetry on physical in-vehicle hardware testbeds.

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